AgroVision

1st Pulak Saha
North South University
Dept: Electrical and Computer Engineering

2nd Shiblu Rahman Shikat
North South University
Dept: Electrical and Computer Engineering

3rd Subrato Chandra Pal
North South University
Dept: Electrical and Computer Engineering

4th Faishal Youshuf
North South University
Dept: Electrical and Computer Engineering

5th Dr. Shahnewaz Siddique
North South University
Associate Professor
Dept: Electrical and Computer Engineering

***Abstract*—Bangladesh is an agriculture dependent country. This report introduces a mobile app called "AgroVision" that uses different smart techniques to improve precision farming in various ways. The application offers a suite of features, including real-time soil detection, crop recommendation, yield prediction, weed detection, and leaf disease identification. Soil analysis utilizes machine learning algorithms like SVM (95%), KNN (94%), CNN (87%), and Random Forest (92%), employing the device's camera for accurate assessments. Crop recommendations and yield predictions are made based on user-input parameters, using models such as SVM (92%), KNN (88%), Decision Tree (99%), and Random Forest (91%). Weed detection, achieved through real-time analysis using YOLOv5 on the device's camera, demonstrates an impressive accuracy of 94%. Leaf disease identification for corn, potato, and rice utilizes SVM (96%) for corn and potato, and Random Forest (92%) for rice. Notably, the best-performing model was converted into a TensorFlow Lite file and seamlessly integrated into the app. "AgroVision" stands as a comprehensive tool for precision farming, encompassing soil analysis, crop recommendations, yield predictions, weed detection, and leaf disease identification.**

***Index Terms*—Precision farming, AgroVision, Machine learning, Mobile application, Soil detection, Crop recommendation, Yield prediction, Weed detection, Leaf disease detection, SVM, Random Forest, Decision Tree, CNN, KNN, AUC, YOLOv5, Cross-validation, Accuracy.**

I. Introduction

Precision farming has become a transformative approach in optimizing agricultural practices, particularly crucial in resource-constrained environments like Bangladesh. The traditional reliance on expert knowledge and experience faces limitations in addressing the complexities of modern agriculture, necessitating a data-driven and technology-enabled solution. In response, "AgroVision," a comprehensive mobile application tailored to the agricultural community in Bangladesh, has been developed. Leveraging cutting-edge machine learning models, AgroVision addresses critical aspects of precision farming, including soil detection, crop recommendation, yield prediction, weed detection, and leaf disease identification. The aim is to empower farmers with precise and efficient solutions, revolutionizing agricultural practices in Bangladesh for increased productivity, improved resource utilization, and sustainable development. Challenges in traditional approaches include limited accessibility to expert knowledge, variability in agricultural conditions, and inadequate disease detection. AgroVision seeks to overcome these challenges by providing accurate soil detection for enhanced soil management, informed crop recommendations for optimized yield potential, real-time weed detection for minimizing crop losses, and timely disease control measures through early detection and identification of leaf diseases. The overarching goal of AgroVision is to contribute to the transformation of agricultural practices in Bangladesh, ensuring sustainable and efficient farming for the betterment of the farming community and the environment. [1]

II. Research Literature Review

S. A. Z. Rahman, K. C. Mitra, and S. M. M. Islam aims for forecasting soil series and offering suitable crop yield suggestions for that particular soil. Six Upazillas in the Khulna region's soil records were used for the research. A variety of machine learning algorithms have been used to test the model. Although Bagged Tree and K-NN exhibit good accuracy, SVM has provided the highest accuracy in classifying soils of all the classifiers which is 94.95%. [2]

P. Malik, S. Sengupta, and J. S. Jadon used machine learning algorithms to analyze on self –obtained dataset, for three crops - tomato, potato, and chili. The crop yield prediction has been done using KNN, Na¨ıve Bayes, and Decision Trees classifier. They find the accuracy of the KNN algorithm 91.179%. The naive Bayes Algorithm predicted 76.426%, and Decision Trees predicted the highest prediction of 95.361%. [3]

F Dang, D Chen, Y Lu, and Z Li present a new dataset (CottoWeedDet12) of weeds that are important to cotton production in the southern United States; it consists of 5648 images of 12 weed classes in cotton fields. They have established YOLO object detectors of all seven versions. They find 88.1 4 % by YOLOv3, 95.2 2 % by YOLOv4 and 68.1 8 % by YOLOv3, 89.7 2 % by Scaled -YOLOv4. [4]

A. V. Panchal, S. C. Patel, K. Bagyalakshmi, P. Kumar, I. R. Khan and M. Soni introduces an image-based plant disease detection system using Convolutional Neural Networks (CNNs). Utilizing a dataset of 87,000 RGB images, trained a VGG16 model with parameter tuning, achieving 93.5% accuracy on validation data. Their study emphasizes the efficacy of deep learning, particularly CNNs, in identifying plant diseases with limited training data. The authors suggest future work involving sensor-based data collection and expanding the model for cure proposals. [5]

III. Methodology

A. System Design

The main components of the AgroVision app are shown in the figure 1's diagram. Its four main purposes are to detect soil, recommend crops and estimate yields, identify weeds, and detect leaf diseases. We can identify weeds, soil, and leaf disease in real time, and we can only do so with a camera feeds. Crop Recommendation analyses user input to identify appropriate crops and calculate yield. UI manages user interactions, while backend fetches data. The app's re a time camera feed analysis and user interaction elements a highlighted in this diagram.

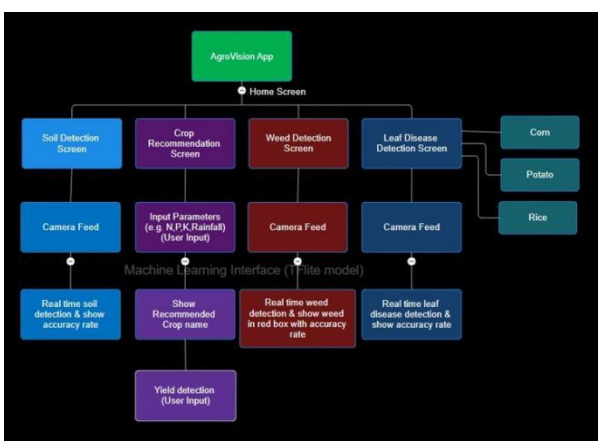

Fig. 1. App's System Diagram

B. Software Components

1) Dataset: The database was collected from an open-source dataset which is completely available in kaggle. We collected 8 types soil image dataset. This type are Yellow Soil, Peat Soil, Laterite Soil, Cinder Soil, Black Soil, Gravel soil, Sand soil, and Silt soil. [6]

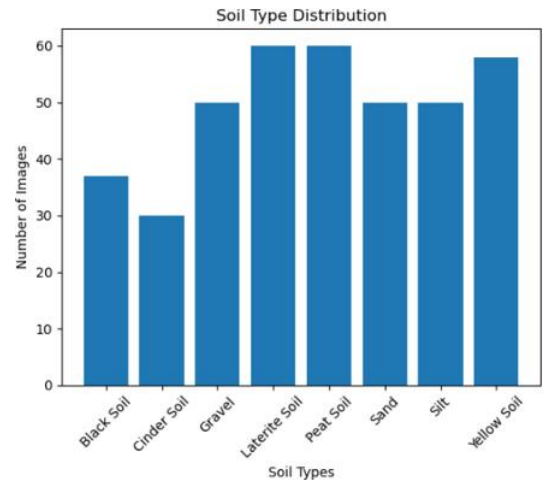

Fig. 2. Soil Distribution

The datasets for crop recommendation we have used here have 8 columns and 2201 rows. After checking the data types and the null values, it was found clean. The feature used for trimming down to the relevant features found the most values column to be "Label." It was discovered that 4.5% of each class in the label is distributed equally. [7]

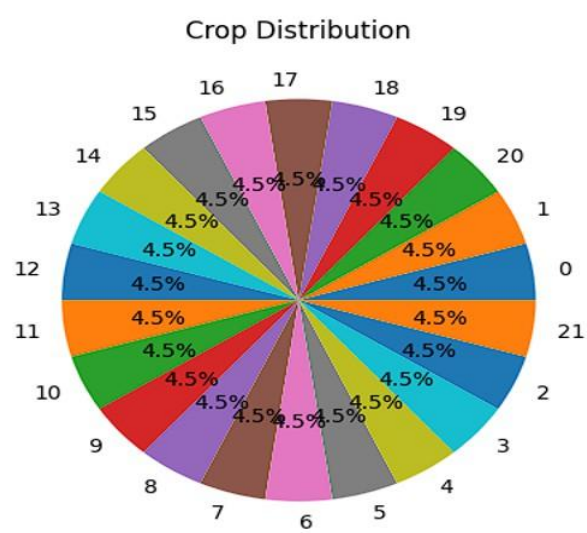

Fig. 3. Crop Distribution

The dataset of weed management integrated with different types of weed images and image labels. We created tabular dataset using labels. The datasets that we have used here have 5 columns and 14920 rows. The feature used for trimming down to the relevant features found the most values column to be "label". [8]

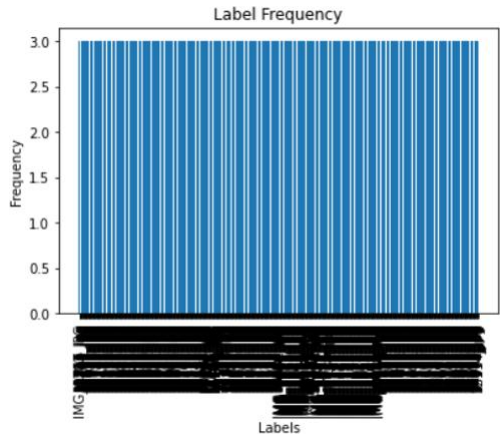

Fig. 4. Bar chart of Weed Images and Labels

We used three different leafs in the leaf disease detection section. We used the Corn, Potato and Rice leafs different disease images as dataset.

The figure 5's pie chart of the distribution of different types of Corn, Potato and Rice leafs leaf disease. The pie chart is titled "Corn leaf disease Type Distribution". The pie chart slices are labeled: [9][10][11]

Corn leafs	Potato leafs	Rice leafs
Blight (27.4%)	Early blight (40.1%)	Bacterial leaf blight (33.3%)
Common Rust (31.2%)	Late blight (34.8%)	Brown spot (33.3%)
Gray Leaf Spot (13.7%)	Other (0%)	Other (0.1%)
Healthy (27.7%)	Healthy (25.1%)	Leaf smut (33.3%)

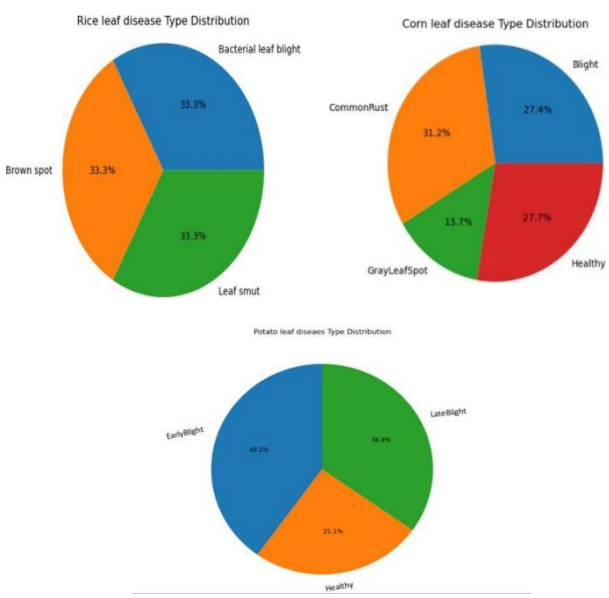

Fig. 5. Corn, Potato, and Rice leaf disease dataset distribution

2) Dataset Preprocessing: The dataset of soil image dataset is divided into two portions, x and y, with y housing all the output columns and x housing the rest. The 'split' command divided the database into two sections, training and testing, with a ratio of 80% for training and testing and 20% for testing, respectively. Since this is the outcome column, all fea- tures were considered during feature selection, excluding the diagnostic features. Algorithms are used to take the features and the resulting column in array format.

3) Preparing Algorithms: Several machine learning, deep learning and computer vision techniques have been used in this paper. We employed a number of techniques to maximize each model's performance prior to implementing it. So in preparation time we follow this steps.

We import some package:

- Define the path to the dataset directory.
- Define the list of soil types, 8 columns of crop recom- mendation dataset, prepare yields dataset, 5 columns of weed label dataset, and 3 types of leafs several types of disease.
- Define the image size for resizing.
- Convert the data and labels to numpy arrays.

Different Algorithms:

The list of used algorithms are

- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)
- Convolutional Neural Networks (CNN)
- Decision Tree
- Random Forest
- You Only Look once (YOLOv5)

C. Software Implementation

1) Front-end Design: The front-end of "AgroVision" is engineered to provide a seamless and intuitive user experi- ence. Developed using the Flutter framework, the application ensures cross-platform compatibility, enabling users on both Android and iOS devices to access its features. Flutter's widget-based approach facilitates the creation of a visually appealing interface, featuring interactive dashboards and real- time visualizations for enhanced user engagement.

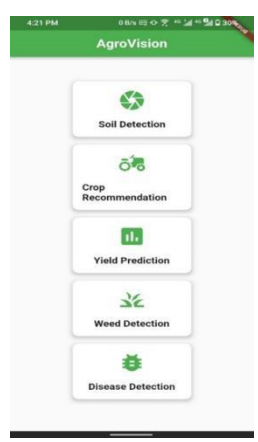

Fig. 6. App's Home Screen

2) Back-end Design: The strategic integration of TensorFlow Lite (TFLite) models into the back-end architecture of the ”AgroVision” mobile application, aiming to provide an efficient and responsive platform for precision agriculture. By seamlessly incorporating optimized machine learning models into the back-end, the study demonstrates how this integration enhances the application’s overall performance and empowers users with real-time insights in the field.

3) Real-Time Analysis Capabilities: The seamless integration of TFLite models into the back-end empowers ”AgroVision” with real-time analysis capabilities. Users can perform complex tasks such as soil analysis, crop recommendations, yield predictions, weed detection, and disease identification directly on their mobile devices. This functionality enhances the application’s responsiveness, enabling farmers to make informed decisions in the field without reliance on external servers.

IV. Investigation/Experiment, Result, Analysis and Discussion

A. Dataset Result, Analysis and Discussion

Soil Detection:

We employed multiple machine learning approaches in our study to predict soil type, including SVM, KNN, CNN, and Random Forest. However, after applying all of these models, we discovered that the most successful model, particularly for soil type prediction, is SVM. SVM offers the best performance and accuracy 95%.

	precision	recall	f1-score	support
0	0.93	1.00	0.96	13
1	0.83	0.91	0.87	11
2	1.00	0.78	0.88	9
3	1.00	1.00	1.00	15
4	1.00	0.83	0.91	12
5	0.85	1.00	0.92	11
6	1.00	1.00	1.00	10
7	1.00	1.00	1.00	12
accuracy			0.95	93
macro avg	0.95	0.94	0.94	93
weighted avg	0.95	0.95	0.95	93

Fig. 7. The precision, recall, f1-score of soil type recognition using SVM model.

Crop Recommendations:

We applied many machine learning approaches in our crop recommendation research, including SVM, KNN, Decision Tree, and Random Forest. However, after testing all of these models, we determined that the most successful model, particularly for crop recommendation, is Decision Tree. Decision Tree provides the greatest performance and accuracy, with a 99% accuracy rate.

	precision	recall	f1-score	support
apple	1.00	1.00	1.00	23
banana	1.00	1.00	1.00	21
blackgram	0.95	1.00	0.98	20
chickpea	1.00	1.00	1.00	26
coconut	1.00	1.00	1.00	27
coffee	1.00	1.00	1.00	17
cotton	1.00	1.00	1.00	17
grapes	1.00	1.00	1.00	14
jute	0.92	0.96	0.94	23
kidneybeans	1.00	1.00	1.00	20
lentil	0.92	1.00	0.96	11
maize	1.00	0.95	0.98	21
mango	1.00	1.00	1.00	19
mothbeans	1.00	0.92	0.96	24
mungbean	1.00	1.00	1.00	19
muskmelon	1.00	1.00	1.00	17
orange	0.93	1.00	0.97	14
papaya	1.00	1.00	1.00	23
pigeonpeas	1.00	1.00	1.00	23
pomegranate	1.00	1.00	1.00	23
rice	0.94	0.89	0.92	19
watermelon	1.00	1.00	1.00	19
accuracy			0.99	440
macro avg	0.98	0.99	0.99	440
weighted avg	0.99	0.99	0.99	440

Fig. 8. The precision, recall, f1-score of Crop Recommendation using Decision Tree model

Yield Detection:

In our agricultural yield prediction research, we evaluated numerous machine learning models such as SVM, KNN, Decision Trees, and Random Forests. Notably, the Random Forests algorithm, in particular, was the most effective, with a staggering accuracy of 99%. This demonstrates Decision Trees’ usefulness in crop yield prediction.

Index(['Crop', 'Hectare', 'Metric Ton'], dtype='object')

	precision	recall	f1-score	support
0	0.99	1.00	0.99	82
1	0.00	0.00	0.00	1
accuracy			0.99	83
macro avg	0.49	0.50	0.50	83
weighted avg	0.98	0.99	0.98	83

Fig. 9. The precision, recall, f1-score of Yield Detection using Random Forest model

Weed Detection:

In our weed identification research, we used YOLOv5. YOLOv5 model displayed extraordinary efficacy, producing the greatest performance and accuracy among the tested approaches, reaching an astonishing 94%. This highlights YOLOv5’s exceptional capabilities in weed recognition, estab- lishing it as the most effective model for precise and efficient weed identification.

	precision	recall	f1-score	support
0	0.67	0.23	0.35	77
1	0.95	0.99	0.97	1117
accuracy			0.94	1194
macro avg	0.81	0.61	0.66	1194
weighted avg	0.93	0.94	0.93	1194

Fig. 10. The precision, recall, f1-score of Weed Detection using Random Forest model

Leaf Disease Detection:

In our project, for leaf disease detection we have used several machine learning methods like SVM, KNN, CNN, and Random Forest. But after implementing all of this model we found that, especially for Rice leaf disease detection most successful one is Random Forest, for Corn and Potato leaf disease detection most successful one is SVM. SVM has the highest performance and accuracy for Corn and Potato leaf disease detection, which is 96% and 95%

	precision	recall	f1-score	support
0	0.98	0.95	0.95	258
1	0.94	0.95	0.97	254
2	0.85	1.00	0.96	104
3	0.86	1.00	0.96	104
accuracy			0.96	838
macro avg	0.90	0.92	0.24	838
weighted avg	0.95	0.96	0.12	838

Fig. 11. The precision, recall, f1-score of Corn Leaf Disease Detection using SVM model

	precision	recall	f1-score	support
0	0.01	0.01	0.01	0
1	0.67	0.67	0.67	3
2	0.01	0.01	0.01	2
3	1.00	0.33	0.50	3
4	0.01	0.01	0.01	2
accuracy			0.95	10
macro avg	0.33	0.20	0.23	10
weighted avg	0.50	0.30	0.35	10

Fig. 12. The precision, recall, f1-score of Potato Leaf Disease Detection using SVM model

On the other hand, Random Forest has the highest performance

and accuracy for Rice leaf disease detection, which is 92%.

	precision	recall	f1-score	support
0	0.67	1.00	0.80	2
1	1.00	0.50	0.67	2
2	1.00	1.00	1.00	2
accuracy			0.92	6
macro avg	0.89	0.83	0.82	6
weighted avg	0.89	0.83	0.82	6

Fig. 13. The precision, recall, f1-score of Rice Leaf Disease Detection using Random Forest model

B. App's Result, Analysis and Discussion

Soil Detection:

In the Soil Detection tab, by using the phone camera tool from this live video feed Agrovision app is showing that the soil type of that land is ' Silt ' and also it is showing the accuracy rate of that prediction.

Fig. 14. Soil Detection interface in the app

Crop Recommendation:

In the Crop Recommendation tab, by inputting the soil nutrition value. If the input of Nitrogen(23), Phosphorous(24), Potassium(35), Temperature(20.2), Humidity(12), pH(9.9), and Rainfall(8). Based on the data Agrovision app is showing that the 'Cotton' will be the best crop for that land.

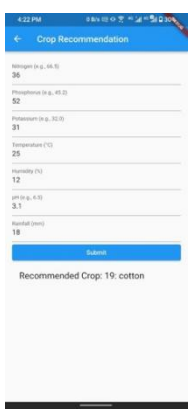

Fig. 15. Crop Recommendation interface in the app

Yield Detection:

We have several crop types listed in our dataset for yield detection. We will have two sections in our dataset. The first is yield prediction, and the second is land area (hectare). For yield prediction, the user has the option to choose a crop from our dataset. After selecting the crop, they must select the land area, which will be measured by hectare. In the land area section, the user will give any integer input. So, after taking the inputs from the user, it will predict the yield. For example, if our crop is maize and our land area is 9 hectares, this will give an estimated yield of 84.6 metric tons.

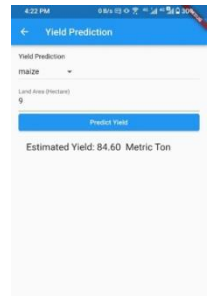

Fig. 16. Yield Detection interface in the app

Weed Detection:

If the user capture a real-time camera feed, it will detect the weed in the field. Here, in picture see that the weeds are predicted by the red box that covers the weed area.

Fig. 17. Weed Detection interface in the app

Leaf Detection:

AgroVision have three leaf diseases detection part. For each crop we have dialog box in our system app.

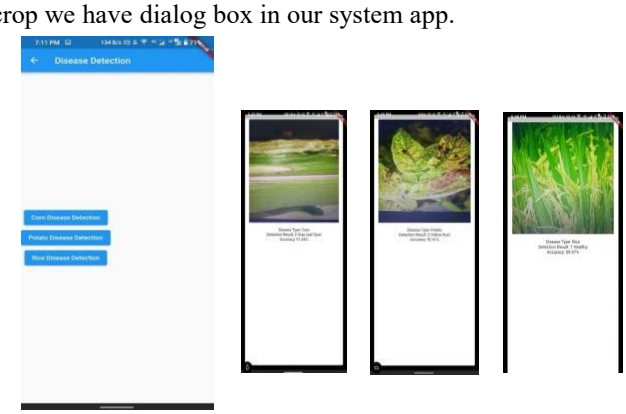

Fig. 18. Leaf Disease Detection interface in the app

When we opened corn leaf disease detection dialog box from app then system scanned corn leaf image and it detect and show diseases gray leaf spot with real time accuracy percentage

After that when we opened potato leaf disease detection dialog box from app then system scanned corn leaf image and it detect and show diseases yellow rust with real time accuracy percentage.

Later on when we opened rice leaf disease detection dialog box from app then system scanned corn leaf image and it detect and show diseases healthy plant with real time accuracy percentage.
Our system app capable for real time leaf disease detection with accuracy using mobile camera with live video feed.

V. Impacts of the Project

A. Impact of this project on societal, health, safety, legal and cultural issues

1) Societal Impact: We had some initial social goals when we thought of building our AgroVision app. For example, firstly, we create a soil type detection solution for farmers to detect the soil type of their land. By seeing soil type, he could make necessary decisions for cultivating the desired crop, which will be perfect for that land. Secondly, we build the crop recommendation solution for farmers to get a brief idea of which crop will be best for that land. We will be able to give maximum production output. Thirdly, we build yield prediction solutions for farmers to get a brief idea about the total yield production of their land. Lastly, we make the weed detection and leaf disease detection solution for farmers, able to detect it early, and they will be able to prevent the spread of leaf disease and weed growing of crops. Therefore, their crop production will not be affected, and they will be able to get the desired output from their land.

2) Health Impact: In terms of health impact, our project aim was to reduce fertilizer and chemicals use on the land.Using our soil type prediction and crop recommendations farmers will be able to reduce the entrance of dangerous chemicals into the soil, protecting both the environment and

3) human health. Furthermore, applying our weed detection and leaf disease detection solutions farmers will able to prevent and treat leaf diseases. Which will adds to overall crop health by minimizing the need for chemical interventions that may pose health hazards. Similarly, employing yield forecasting, increases crop output and reduces dependency on potentially toxic chemicals.

4) Safety Impact: Our AgroVision app depends on real-life data. As our land's climate and condition constantly changes, we have to continually update our data collection so that our model can accurately predict soil type, crop recommendations, yield, weed detection, and leaf disease detection. Otherwise, the inaccurate prediction from our app will misguide our farmers, leading us to catastrophes like country food shortages and famine.

5) Legal Impact: We have to develop clear policies for appropriate agricultural use, which will resolve legal issues with forecast errors and protect intellectual property. We must have clear and concise disclosures essential for managing legal risks associated with projected variances. Besides, we must have patents and intellectual property protection methods for owning agricultural technologies.

6) Cultural Impact: Most of the people in our country still depend on agriculture. Sadly our farmers still follows the traditional farming approach. Because of that reason we are not able to utilize our land potential fully. For example: according to google data Bangladesh produce wheat 3.37 ton/ha, whereas USA produce wheat 7.8 ton/ha. Our goal is combine traditional farming knowledge with technology while ensuring that technical solutions adhere to cultural standards.

B. Impact of this project on environment and sustainability

We aim to simplify agriculture for everyday people by effectively applying our approach. Farmers will also be able to boost crop yield and use agricultural land more efficiently by following suitable suggestions from our application. We will be able to forecast the country's entire crop production and meet our people's food needs. We will also be able to lower the amount of fertilizer we use on our agricultural land. We will be able to identify and diagnose agrarian disease. At last, we can utilize our land sustainably and lessen our negative effects on the ecosystem.

VI. Project Planning

This project, which was began on January 25, 2023, and was completed on November 27, 2023, successfully developed a system for detecting soil type, recommending crops, identifying weeds, predicting crop yield, and recognizing leaf diseases. The project progressed smoothly through its phases: planning, proposal and data collection, soil detection and crop recommendation, weed detection, 1st progress report, crop yield prediction, leaf disease detection, prototyping, testing and evaluation, and final report. The project's completion marks will be a significant milestone in the field of precision agriculture.

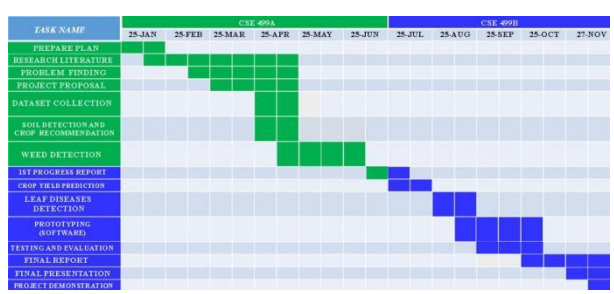

Fig. 19. Project Planning Gantt chart

VII. Conclusion

A. Summary

AgroVision is an innovative agriculture-focused mobile app designed to assist farmers and individuals in the agricultural sector. The app's features include soil detection through the device's camera, employing a machine learning model for real- time identification of soil types with accuracy percentages. It offers crop recommendations based on user-input parameters such as soil conditions, temperature, humidity, pH levels, and rainfall, and predicts crop yield by considering area and crop name. The app also facilitates real-time weed detection through the camera, aiding farmers in timely intervention. Additionally, AgroVision enables users to diagnose diseases in crop leaves, such as corn, potato, and rice, using a machine learning model for accurate identification with associated ac- curacy percentages. This comprehensive set of features equips farmers with valuable tools for soil analysis, crop management, and disease prevention, contributing to more informed and efficient agricultural practices.

B. Future Improvement

In the future, AgroVision could explore further enhance- ments to solidify its position as an indispensable tool for the agricultural community. Integration with real-time weather data and satellite imagery could offer more precise insights into environmental conditions, enabling farmers to make even more informed decisions.

VIII. References

[1] 1. Sharma, A., Jain, A., Gupta, P., Chowdary, V. (2020). Machine

learning applications for precision agriculture: A comprehensive review.

IEEE Access, 9, 4843-4873.

[2] 2. Rahman, S. A. Z., Mitra, K. C., Islam, S. M. (2018, December).

Soil classification using machine learning methods and crop suggestion

based on soil series. In 2018 21st International Conference of Computer

and Information Technology (ICCIT) (pp. 1-4). IEEE.

[3] 3. Malik, P., Sengupta, S., Jadon, J. S. (2021, January). Comparative

analysis of soil properties to predict fertility and crop yield using

machine learning algorithms. In 2021 11th International Conference on

Cloud Computing, Data Science Engineering (Confluence) (pp. 1004-

1007). IEEE.

[4] 4. Dang, F., Chen, D., Lu, Y., Li, Z. (2022). YOLOWeeds: A novel

benchmark of YOLO object detectors for weed detection in cotton

production systems. Computers and Electronics in Agriculture, July.

[5] 5. A. V. Panchal, S. C. Patel, K. Bagyalakshmi, P. Kumar, I. R. Khan and

M. Soni, M. (2023). Image-based plant diseases detection using deep

learning. Materials Today: Proceedings, 80, 3500-3506.

[6] 6. Soil Types. (n.d.). Soil Types — Kaggle.

https:///datasets/prasanshasatpathy/soil-types

[7] 7. Crop analysis and prediction. (n.d.). Crop Analysis and Pre

diction — Kaggle. https://www.kaggle.com/code/theeyeschico/crop

analysis-and-prediction

[8] 8. DeepWeeds. (n.d.). DeepWeeds — Kag

gle. https:///datasets/imsparsh/deepweeds

sklearn.svm.SVC. (n.d.). Scikit-learn.

https://scikitlearn/stable/modules/generated/sklearn.svm.SVC.html

[9] 9. Corn Leaf Diseases: Corn or Maize Leaf

Disease Dataset. (2020, November 11). Kaggle.

https://www.kaggle.com/datasets/smaranjitghose/corn-or-maize-leaf

disease-dataset[10] 10. Potato Leaf Diseases: Potato Disease Leaf

Dataset(PLD). (2021, August 31). Kaggle.

https://www.kaggle.com/datasets/rizwan123456789/potato-disease

leaf-datasetpld

[11] 1. Rice Leaf Diseases: Rice Leaf Diseases Dataset. (2020, February 21).

Kaggle. https://www.kaggle.com/datasets/vbookshelf/rice-leaf-diseases